Agentic AI for Earth Observation

19–21 October 2026 · Berlin, Germany

Day 1 — Monday, 19 October 2026

Time	Foyer & Wandelhalle Ground Floor & 1st Floor	Lecture Hall 2nd Floor	Expertisium Ground Floor	B. von Langenbeck 1st Floor	Werner-Körte 1st Floor	Rudolf-Virchow 2nd Floor
12:00 – 13:00	Registration & Welcome Lunch					
13:00 – 13:45		Opening Remarks				
13:45 – 14:30		Keynote 1				
14:30 – 15:00	Coffee Break | 4 Demos Foyer: Demos 13,14 Wanderhalle: Demos 11,15		Coffee Break | 2 Demos Demos 12,17		Featured Demo 1	Featured Demo 2
15:00 – 16:00		Oral Session 1 – Agentic Decision Intelligence and Earth Action				
16:00 – 16:45		Panel Discussion 1				
17:00 – 18:40		Thematic Discussion 17:00–17:30		Poster Session 1 17:00–18:30	**17:00–17:25** Featured Demo 3 **17:25–17:50** Featured Demo 5 **17:50–18:15** Featured Demo 7 **18:15–18:40** Featured Demo 9	**17:00–17:25** Featured Demo 4 **17:25–17:50** Featured Demo 6 **17:50–18:15** Featured Demo 8 **18:15–18:40** Featured Demo 10
18:45 – 20:45	Welcome Reception					

Day 2 — Tuesday, 20 October 2026

Time	Foyer & Wandelhalle Ground Floor & 1st Floor	Lecture Hall 2nd Floor	Expertisium Ground Floor	B. von Langenbeck 1st Floor	Werner-Körte 1st Floor	Rudolf-Virchow 2nd Floor
08:00 – 08:45	Welcome Coffee					
08:45 – 09:30		Keynote 2				
09:30 – 10:15		Keynote 3				
10:15 – 10:45	Coffee Break | 4 Demos Foyer: Demos 3,5 Wanderhalle: Demos 20,21		Coffee Break | 2 Demos Demos 8,22		Featured Demo 11	Featured Demo 12
10:45 – 11:45		Oral Session 2 – Autonomous EO Systems, Space Operations and Cognitive Computing in Space				
11:45 – 12:30		Panel Discussion 2				
12:30 – 12:45		Group Photo Shoot				
12:45 – 14:00	Hosted Lunch					
14:00 – 14:30		Keynote 4				
14:30 – 15:45		Oral Session 3 – Design, Development and Evaluation of AI Agents				
15:45 – 16:15	Coffee Break | 4 Demos Foyer: Demos 1,7 Wanderhalle: Demos 2,4		Coffee Break | 2 Demos Demos 6,9		Featured Demo 13	Featured Demo 14
16:15 – 17:15		Technical deep dives: Building with Mistral Open Agentic AI tools: From Scientific Discovery to Operational Services				
17:15 – 17:45		Keynote 5				
17:30 – 19:10			Poster Session 2 17:45–19:00		**17:45–18:10** Featured Demo 15 **18:10–18:35** Featured Demo 17 **18:35–19:00** Featured Demo 19	**17:45–18:10** Featured Demo 16 **18:10–18:35** Featured Demo 18 **18:35–19:00** Featured Demo 20

Day 3 — Wednesday, 21 October 2026						
Time	**Foyer & Wandelhalle** Ground Floor & 1st Floor	**Lecture Hall** 2nd Floor	**Expertisium** Ground Floor	**B. von Langenbeck** 1st Floor	**Werner-Körte** 1st Floor	**Rudolf-Virchow** 2nd Floor
08:00 – 09:00	Welcome Coffee					
09:00 – 09:45		Keynote 6				
09:45 – 10:30		Keynote 7				
10:30 – 11:00	Coffee Break | 2 Demos Foyer: Demos 18,19		Coffee Break | 2 Demos Demos 10,16		Featured Demo 21	Featured Demo 22
11:00 – 12:00		Oral Session 4 – Multi-Modal Large Models, Datasets and Agentic Earth Intelligence				
12:00 – 12:45		Closing Ceremony				
12:45 – 14:00	Hosted Lunch					
14:00 – 15:30			**Track 1: Building and Operating EO Agents** Building Agentic Earth Observation Workflows with EVE Workbench	**Track 2: Trustworthy and Scientific Agentic AI** How Do You Know Your Agent Works? Evaluating Earth Observation Agents with Existing Benchmarks		**Track 3: Foundation Models, Embeddings and Lightweight AI (Part 1)** Small Models, Big Maps – Local Geospatial Workflows with Fine-tuned SLMs
15:30 – 16:00	Coffee Break					
16:00 – 17:30			**Track 1: Building and Operating EO Agents** Agentic Analysis of Earth Observation for Utility Intelligence	**Track 2: Trustworthy and Scientific Agentic AI** From Vibe Coding to Verifiable Scientific Discovery: Agentic Workflows for Earth Observation		**Track 3: Foundation Models, Embeddings and Lightweight AI (Part 2)** Small Models, Big Maps – Local Geospatial Workflows with Fine-tuned SLMs

Detailed Program

Day 1 - Monday, October 19, 2026

12:00 – 13:00 Registration & Welcome Lunch

13:00 – 13:45 Opening Remarks

Speakers:

Prof. Dr. Fatma Deniz, President of TU Berlin

Prof. Dr. Volker Markl, Director of BIFOLD

Dr. Claudio Iacopino, Head of ϕ-lab Explore Office, European Space Agency

Prof. Dr. Begüm Demir, Deputy Director of BIFOLD

Dr. Nicolas Longépé, EO Data Scientist at ϕ-lab Explore Office, European Space Agency

13:45 – 14:30 Keynote 1

Agentic AI for Earth Observation and Climate Resilience
Presented by Prof. Dr. Salman Khan, Mohamed bin Zayed University of Artificial Intelligence

14:30 – 15:00 Coffee Break & Demos

Foyer

Coffee Break

Demo 13: From Chat to Copernicus: Agent-Based Composition of Reproducible openEO Workflows on CDSE – VITO
Presented by Hans Vanrompay, Victor Verhaert, Mathias Michielsen

Demo 14: SAI4EO: From Conversational Remote-Sensing Tool Use to Agentic, Artifact-Grounded Geospatial Intelligence – Planetek Italia
Presented by Nicolò Taggio

Wanderhalle

Coffee Break

Demo 11: EVE Workbench Live: A Multi-Agent Debate on Berlin's Climate Stress – Frank Reply
Presented by Daniel Fitzpatrick, Maria Müller

Demo 15: Portable Agentic openEO: From Natural Language to Executable Earth Observation Workflows – EURAC Research Italy
Presented by Suriyah Dhinakaran

Expertisium

Coffee Break

Demo 12: DEA Next: Making Earth Intelligence Conversational through Agentic AI – Alia Space Systems
Presented by Arturo Montieri

Demo 17: terraAI: Conversational Earth Observation Analysis – Vilnius University
Presented by Evgenij Shapovalov, Artiom Hovhannisyan, Valentas Gružauskas

Werner-Körte

Featured Demo 1: Earthward: A Human-in-the-Loop Architecture for Traceable, Explainable Agentic EO Workflows – Earthward
Presented by Sudhakaran Chithamanan, Aravindh S. Subramanian

Rudolf-Virchow

Featured Demo 2: Saferplaces Agentic AI - An Agentic Geospatial Copilot for Flood Risk Intelligence and Map-Based Decision Support – GECOsistema SRL
Presented by Tommaso Redaelli

15:00 – 16:00 Oral Session 1 - Agentic Decision Intelligence and Earth Action *Moderated by Chair to be filled*

This session focuses on agents that transform EO observations into actionable decisions. This session illustrates the evolution: from Observation, analysis, prioritization, decision support and Earth Action.

15:00 – 15:12 A Decision-Driven Hazard-Agnostic Agentic AI Framework for Earth Observation: Demonstration through a Flood Response Use Case
Presented by Firoozeh Foroozan and Carlos Pino Padilla, Indra Space

15:12 – 15:24 Multi-Agent Early Extreme Event Detection From IoT to Earth Observation Tasking
Presented by Arnaud Le Carvennec and Axel Gruyez, CS Group

15:24 – 15:36 Wildfire Lifecycle Monitoring from Earth Observation
Presented by Rishi Madhok and Fuxun Yu, TerraByte

15:36 – 15:48 Agentic Orchestration of Earth Observation Foundation Models via the Model Context Protocol
Presented by Pauline Delporte and Marie Leconte and Arnaud Le Carvennec, CS Group, and Nicolas Longépé, ESA Φ-Lab

15:48 – 16:00 TerraScope: Pixel-Grounded Visual Reasoning for Earth Observation
Presented by Yan Shu, University of Trento

16:00 – 16:45 Panel Discussion 1 – EO in the age of Agentic AI: From Insight to Business *Moderated by Chiara Cocchiara, ESA Φ-lab*

From data to insight, and from insight to action: The panel will explore how Agentic AI could fundamentally reshape the EO value chain, from sensing and data generation to analysis, decision-making and business impact. Bringing together perspectives from research, EO technology and enterprise, the discussion will examine how AI agents could be a key element of the full data chain, combine EO with diverse sources of knowledge, and transform increasingly rich Earth observations into decision-ready intelligence. With applications ranging from insurance and risk management to emerging EO services and markets, the panel will address the key question: How do we move from technological potential to real-world impact and sustainable business value?

Panelists:

Julia Gottfriedsen, Chief Technical Officer at ConstelIR

Marcin Detyniecki, Group Chief Data Scientist & Head of Research & Advanced AI at AXA

Andres Camero, Co-Head of Department EO Data Science at German Aerospace Center (DLR)

Linda Sheard, VP, Analytics and Agentic Products at Xoople

17:00 – 18:40 Thematic Discussion, Poster Session 1 & Demos

17:00 – 17:45 Lecture Hall: Thematic Discussion - Subway Takes: Agentic AI Edition, moderated by Eva Gmelich Meijling and Riccardo D'Ercole - ESA Φ-lab

Join this guided, engaging discussion where participants share and debate strong opinions, concerns, and unpopular takes on Agentic AI for Earth Observation. Come with a take of your own, or respond to one raised by someone else. The goal is not consensus, but a better collective understanding of where agentic approaches genuinely add value in EO.

17:00 – 18:30 B. von Langenbeck: Poster Session 1

Fitness for Purpose as an Agentic Task: Domain Encoded Reasoning for EO Data Selection in Regulated Environmental Work
Presented by Nahid Alemi Kermani, Aniterra Pty Ltd

SENSOR2RAG: Multimodal Sensor-to-RAG Architecture for Site-Specific Decision Support in Natural Capital Risk Management
Presented by Jacopo Soldateschi, The Landbanking Group GmbH

Beyond Task Success: Domain-Grounded Evaluation of Earth Observation Agents in Environmental Risk Workflows
Presented by Alexandros Oikonomidis, Centre for Research and Technology Hellas (CERTH) and National and Technical University of Athens (NTUA)

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings
Presented by Jaya Kumari, Vortx AI

CLUES: A Comprehensive Workflow for Integrating Geospatial Data in Biomedical Research
Presented by Marcel Jentsch, Center of Digital Health BIH@Charite

Virtual Support for Earth Observation (VISEO)
Presented by Romain Sixt and Killian Perriot and Dorian Voydie, THALES

GEODES Assistant
Presented by Killian Perriot, THALES

From Isolated Models to Agentic Orchestration: A Ground Segment Multi-Agent Framework for Coastal and In-Land Water Resilience
Presented by Roger Escolà and Mariona Badenas-Agusti, isardSAT

Thematic Datacubes On-Demand (THEDE): An Agentic AI Framework for On-Demand EO Data Dissemination Using Large Language Models
Presented by Edoardo Kimani Bellotto, Sistema GmbH

EIS: An Open-Source Geo-Agent Framework Democratizing Environmental Knowledge and Collaboration
Presented by Matteo Mattiuzzi, European Environment Agency

AI Agentic Development of xAI for Earth Observation: Building SegExplainer
Presented by Yulia Grushetskaya and Mike Sips, GFZ Helmholtz Centre for Geosciences

Ask, Detect, Explain: An End-to-End Agentic System for Satellite Change Detection
Presented by Bartłomiej Ostrowski and Marcin Kluczek and Bartosz Augustyn and Jędrzej S. Bojanowski, CloudFerro S.A.

Agentic Interaction between Raster Flood Observations and GIS Features for Crisis Situational Awareness
Presented by Jente Bosmans, VITO

LATINSIGHT: A React-Based Multi-Agent System for Urban EO Analytics
Presented by Edoardo Schettino and Paolo De Piano, Latitudo 40 S.r.l.

ECHOSAT: Physically Consistent Spatio-Temporal Canopy Height Mapping as an Enabling Layer for Agentic Earth Observation
Presented by Berkant Turan, Zuse Institute Berlin

Engineering a Trustworthy NASA Earth Science Discovery Agent Using the CARE Methodology
Presented by Lillianne Thomas, Development SEED

A Simulation Study of LLM-Based Onboard Orchestration for Coordinated Earth-Observation Constellations
Presented by Christopher Castro Traba

Agentic Ground Segment: How Generative AI Redesigns Data Discovery and Processing, Ordering, and Operations
Presented by Filippo Bianchini, Telespazio S.p.A.

Hierarchical System with Agentic Refinement for Large-Scale Earth Observation Archives
Presented by Omjyoti Dutta and Joao Vitorino, GMV S.L.

AI-Driven ASAP Country Assessments
Presented by Filip Sabo, European Commission, Joint Research Centre (JRC)

From Earth Observation Data to Decision: An Agentic Architecture for Disaster Resilience
Presented by Bartosz Augustyn and Marcin Kluczek, CloudFerro S.A.

Advancing Disaster Resilience through Agentic AI Integration in the EO-PIN Suite
Presented by Guy Schumann, RSS-Hydro

EO Agents in Actions - Lessons Learned from Building Products Around Analysis Agents
Presented by Lillianne Thomas and Drishtie Patel, Development SEED

17:00 – 18:40 Werner-Körte

17:00 – 17:25 Featured Demo 3: Virtual Support for Earth Observation (VISEO): Agentic AI for Earth Observation Discovery – Thales Services Toulouse
Presented by Dorian Voydie, Killian Perriot, Romain Sixt

17:25 – 17:50 Featured Demo 5: WRI's Global Nature Watch: Geospatial AI in Plain Language – Development Seed Lisbon
Presented by Daniel Wiesmann, Soumya Ranjan, Leo Thomas

17:50 – 18:15 Featured Demo 7: From Pixels to Risk: Agentic Earth Observation for Explainable Climate Intelligence – Independent
Presented by Claudio Giorgio Giancaterino

18:15 – 18:40 Featured Demo 9: Axis Spatial Migration Engine: From Fragmented EO/GIS Projects to Verified Pipelines – Axis Spatial
Presented by Najah Pokkiri

17:00 – 18:40 Rudolf-Virchow

17:00 – 17:25 Featured Demo 4: An Agentic Digital Twin Framework for Flood Risk Assessment: Integrating Complex Workflows with Generative AI: From Satellite Acquisition to Risk Management – CS Group
Presented by Pauline Delporte, Yoann Steux

17:25 – 17:50 Featured Demo 6: Max-ICS V2.0: Toward an Agentic EO Orchestration Platform – EarthLab Luxembourg S.A.
Presented by Mohamed Boukhebouze

17:50 – 18:15 Featured Demo 8: GEONOVA: An Agentic EO Copilot for Geospatial Embeddings and Downstream Model Adaptation – Geonova P.C.
Presented by George Arvanitakis, Georgios Tsimiklis, Elisavet Parselia, Eleftheria Tetoula Tsonga, Dionysos Kampouris, Christos Tsekes

18:15 – 18:40 Featured Demo 10: BlazerOps: Reinforcement Learning Environment for Accelerated Spatial ML Development – hyades.space
Presented by Ashin Alex, Harshil Magan

18:45 – 20:45 Welcome Reception

Langenbeck-Virchow

Detailed Program

Day 2 - Tuesday, October 20, 2026

08:00 – 08:45 Welcome Coffee

Foyer & Wandelhalle

08:45 – 09:30 Keynote 2

(Tentative) The Rise of Agentic AI: Ecosystem, Trends and Implications for Science and Industry
Presented by Mistral AI

09:30 – 10:15 Keynote 3

FAME – Traditional and Agentic AI in Space
Presented by Dr. Steve Chien, Jet Propulsion Laboratory (JPL), NASA

10:15 – 10:45 Coffee Break & Demos

Foyer

Coffee Break

Demo 3: Virtual Support for Earth Observation (VISEO): Agentic AI for Earth Observation Discovery – Thales Services Toulouse
Presented by Dorian Voydie, Killian Perriot, Romain Sixt

Demo 5: WRI's Global Nature Watch: Geospatial AI in Plain Language – Development Seed Lisbon
Presented by Daniel Wiesmann, Soumya Ranjan, Leo Thomas

Wandelhalle

Coffee Break

Demo 20: FLAP-0: An Interactive Platform for Agentic Earth Observation – Barcelona Supercomputing Center and Universitat Politècnica de Catalunya
Presented by Leonidas Kosmidis, Iker Fernandez Andonegui, Marc Solé i Bonet

Demo 21: A Live Tour of EVE: An Agentic EO Platform Built on an Open MCP Tool Registry – Pi School
Presented by Àlex R. Atrio, Antonio Lopez, Jino Rohit, Emanuele Micieli

Expertisium

Coffee Break

Demo 8: GEONOVA: An Agentic EO Copilot for Geospatial Embeddings and Downstream Model Adaptation – Geonova P.C.
Presented by George Arvanitakis, Georgios Tsimiklis, Elisavet Parselia, Eleftheria Tetoula Tsonga, Dionysos Kampouris, Christos Tsekes

Demo 22: Agentic AI-Enabled Climate Downscaling with NASA's Prithvi WxC Foundation Model – JPL/NASA, University of Utah
Presented by Hugo Lee, Aashish Panta

Werner-Körte
Featured Demo 11: EVE Workbench Live: A Multi-Agent Debate on Berlin's Climate Stress – Frank Reply
Presented by Daniel Fitzpatrick, Maria Müller
Rudolf-Virchow
Featured Demo 12: DEA Next: Making Earth Intelligence Conversational through Agentic AI – Alia Space Systems
Presented by Arturo Montieri

10:45 – 11:45 Oral Session 2 – Autonomous EO Systems, Space Operations and Cognitive Computing in Space *Moderated by Chair to be filled*

This session represents the most visionary part of the programme, around Cognitive Cloud Computing in Space (3CS), EO Virtual Constellations and autonomous missions.
"How do we move agentic EO from the cloud to autonomous constellations operating in orbit?"

10:45 – 10:57 Agentic Orchestration of Satellite Tasking and Data Retrieval in Earth Observation Virtual Constellations
Presented by Marie Leconte and Arnaud Le Carvennec and Pauline Delporte, CS Group

10:57 – 11:09 Autonomous Agentic Command and Control for Satellite Constellations: An AI-Accelerated Engineering Demonstration
Presented by Alessio Auddino and Francesco Corallo, Thales Alenia Space Italia

11:09 – 11:21 Towards Cognitive Cloud Computing in Space: AI-eXpress as an Orbital Testbed for Agentic Earth Observation
Presented by Vito Fortunato, Planetek Italia

11:21 – 11:33 Agentic Mission Computing for Autonomous Earth Observation
Presented by Dave Ojika and Nitesh Bakhati and Srinidhi Magesh, Flapmax, and Leonidas Kosmidis and Iker Fernandez Andonegui and Marc Solè i Bonet and Eric Rufart Blasco, Barcelona Supercomputing Center and Universitat Politècnica de Catalunya, and Projjal Gupta and Herman Lam, University of Florida, and Aidin Osucht, University of California, and Eric Lim and Isabella Pier and Ben Cragin, University of Southern California

11:33 – 11:45 Closing the Reasoning Latency Gap: FPGA-Accelerated LLM Agents for Onboard Supervision of Earth Observation Acquisition
Presented by Winfred Ooh-Azlin, Technische Universität Berlin, and Pao Ying Heng, Royal Astronomical Society

11:45 – 12:30 Panel Discussion 2 *Moderated by Chair to be filled*

To be filled...

Panelists:

Person...

Person...

Person...

12:30 – 12:45 Group Photo Shoot

On the stairs and in the Foyer

12:45 – 14:00 Hosted Lunch

Langenbeck-Virchow-Haus

Food and drinks will be provided by the workshop free of charge

14:00 – 14:30 Keynote 4

Failure Mode in Agentic Reasoning
Presented by Prof. Dr. Manling Li, Northwestern University & Amazon Scholar

14:30 – 15:45 Oral Session 3 – Design, Development and Evaluation of AI Agents *Moderated by Chair to be filled*

This session addresses perhaps the most critical scientific question emerging from the workshop: "How should EO agents be built, evaluated, validated and trusted?".
This session is likely to generate discussion about evaluation, reproducibility, hallucinations, validation, governance, or trust.

16:15 – 16:27 Collaborative Agent Reasoning Engineering (CARE): A Structured, Context-Engineering Methodology for Systematically Designing Traceable and Maintainable AI Agents for Science
Presented by Rohit Sisir Sahoo, The University of Alabama in Huntsville

16:27 – 16:39 Evaluating EVE, an EO-Specialized LLM, in Agentic Earth Observation Systems
Presented by Riccardo D'Ercole and Eva Gmelich Meijling, ESA ϕ-lab

16:39 – 16:51 EO Agent Arena: From Fragmented Benchmarks to Reproducible Evaluation of Agentic Earth Observation Systems
Presented by Najah Pokkiri, Axis Spatial

16:51 – 17:03 Behavioural Self-Monitoring for Autonomous Earth Observation Agents: The Attribute Reliability Monitor (ARM)
Presented by Alejandro Mousist, Thales Alenia Space

17:03 – 17:15 Balancing Accessibility and Scientific Validity in the Era of AI Agents and Climate Change
Presented by Elena Di Medio and Laia Romero and Guillermo Grau, Lobelia Earth

17:15 – 17:27 From Tools to Ecosystem: EVE's Community-driven MCP Registry for an Agentic Platform for Earth Intelligence
Presented by Àlex R. Atrio and Antonio Lopez and Jino Rohit and Emanuele Micieli, Pi School

15:45 – 16:15 Coffee Break & Demos

Foyer

Coffee Break

Demo 1: Earthward: A Human-in-the-Loop Architecture for Traceable, Explainable Agentic EO Workflows – Earthward
Presented by Sudhakaran Chithamanan, Aravindh S. Subramanian

Demo 7: From Pixels to Risk: Agentic Earth Observation for Explainable Climate Intelligence – Independent
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Wandelhalle

Coffee Break

Demo 2: Saferplaces Agentic AI - An Agentic Geospatial Copilot for Flood Risk Intelligence and Map-Based Decision Support – GECOsistema SRL
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Demo 4: An Agentic Digital Twin Framework for Flood Risk Assessment: Integrating Complex Workflows with Generative AI: From Satellite Acquisition to Risk Management – CS Group
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Expertisium

Coffee Break

Demo 6: Max-ICS V2.0: Toward an Agentic EO Orchestration Platform – EarthLab Luxembourg S.A.
Presented by Mohamed Boukhebouze

Demo 9: Axis Spatial Migration Engine: From Fragmented EO/GIS Projects to Verified Pipelines – Axis Spatial
Presented by Najah Pokkiri

Werner-Körte

Featured Demo 13: From Chat to Copernicus: Agent-Based Composition of Reproducible openEO Workflows on CDSE – VITO
Presented by Hans Vanrompay, Victor Verhaert, Mathias Michielsen

Rudolf-Virchow

Featured Demo 14: SAI4EO: From Conversational Remote-Sensing Tool Use to Agentic, Artifact-Grounded Geospatial Intelligence – Planetek Italia
Presented by Nicolò Taggio

16:15 – 17:15 Technical Deep Dive

Building with Mistral Open Agentic AI tools: From Scientific Discovery to Operational Services
Presented by Mistral AI

17:15 – 17:45 Keynote 5

Harnessing the Collective Intelligence of AI Agents for Discoveries
Presented by Prof. Dr. James Zou, Standford University

17:45 – 19:00 Thematic Discussion, Poster Session 2 & Demos

17:45 – 19:00 B. von Langenbeck: Poster Session 2

Autonomous Geospatial Intelligence: An Agentic Workflow for End-to-End Feature Interpretation And Large-Scale Scene Monitoring
Presented by Marie-Charlotte Teulières, Airbus Defense and Space Geo

EOMAS – A Multi Agent System-Backed Assistant for Earth Observation
Presented by Hans Meine, Fraunhofer Institute for Digital Medicine MEVIS and OHB Digital Services

An Interactive Multi-Agent System for User-Driven And Context-Aware Hyperspectral Image Compression
Presented by Tristan Reimann and Maxim Smirnov and Viktor Tevosyan, Technische Universität Berlin and BIFOLD

Towards Earth Observation-Based Siting of Sponge-City Infrastructure in East African Cities: An Open-Data Framework
Presented by Karl A. W. Amolo, Urban Canopy Kenya and East African Sponge Cities Initiative (EASCI)

AI-Driven End-To-End (E2E) Monitoring in the Context Of The ESA Earth Observation Operations Framework Copernicus Space Component (EOF-CSC)
Presented by Caroline Ball, Capgemini (France)

From Perception to Priorities: An Agentic Pipeline for Multi-Hazard Triage and Explainable Resource Allocation in Earth Observation
Presented by Alessandro Princi, RINA Consulting S.p.A

Learned Satellite Embeddings as Covariates for Global Digital Soil Mapping: A Multi-Model, Multi-Resolution Evaluation Developed with AI Coding Agents
Presented by Laura Poggio, ISRIC - World Soil Information

Specialized Geospatial Foundation Models for Agentic Earth Observation: Lessons from Spheer FM
Presented by Mark Boer and Jordy de Lange, Spheer

Multi-Satellite Task Orchestration for Wildfire Monitoring with On-Board Visual-Language Models
Presented by Riccardo D'Ercole, ESA Φ-lab and Ana-Maria Pelin, Φ-lab and TU Delft

Provenance-First Geospatial Composition: A Scaffold-Based Builder as a Trust-Bearing Counterpoint to Agentic Autonomy in Earth Observation
Presented by Sudhakaran Chithamanan, Earthward, 33 Blueprints Collective and Aravindh S. Subramanian, Technical University of Munich, Earthward

Beyond the State of the Art in Geospatial Foundation Models: A Diagnostic Comparison of TerraMind and THOR
Presented by Frederick Schindlegger, Bertelsmann Foundation

Agentic AI as an Auditable Co-Scientist for Earth-Observation-Based Irrigation Mapping and Groundwater Accounting
Presented by Peter ReVelle, Desert Research Institute (DRI)

SeismicAgent: Benchmarking Evidence-Conflict Reasoning for Post-Earthquake Building-Damage Assessment
Presented by Fatma Elik, Istanbul Metropolitan Municipality, Earthquake and Ground Investigation Directorate

Towards Agentic AI Workflows for Earth Observation-Based Air Quality Studies
Presented by Alberto Vavassori and Yasil Yordanov, Politecnico di Milano

Enabling Adaptive Agentic Earth Observation across the Space–Ground–Cloud Continuum
Presented by Cecilia Calavaro, Department of Civil Engineering and Computer Science Engineering, University of Rome Tor Vergata

Manteo AI: Agentic Geospatial Intelligence for Earth Observation
Presented by Ioannis Prapas, manteo.ai

Airbus Geo Explore: Leveraging Semantic Search APIs for LLM Agentic Workflows
Presented by Fahd Benatia, Airbus DS

Aperture: Semantic Change Understanding and Grounded GeoAI Reasoning via SAR-Optical-Language Alignment
Presented by Magdalena Kaiser and Paul Weinmann, LiveEO

FlyPix AI: Agentic Vision-Language System
Presented by Ivan Tankoyeu, Flypix AI

GeoGuard: An Agentic Guardrails and Validation Framework for Geospatial AI
Presented by Rohit Sisir Sahoo, The University of Alabama in Huntsville

17:45 – 19:00 Werner-Körte

17:45 – 18:10 Featured Demo 15: Portable Agentic openEO: From Natural Language to Executable Earth Observation Workflows – EURAC Research Italy
Presented by Suriyah Dhinakaran

18:10 – 18:35 Featured Demo 17: terraAI: Conversational Earth Observation Analysis – Vilnius University
Presented by Evgenij Shapovalov, Artiom Hovhannisyan, Valentas Gružauskas

18:35 – 19:00 Featured Demo 19: Operating an EO Constellation with AI Agents – Aistralys
Presented by Alexandru Moloceniuc

17:45 – 19:00 Rudolf-Virchow

17:45 – 18:10 Featured Demo 16: AKD Labs: An End-to-End Agent Co-Design Studio – The University of Alabama in Huntsville, NASA Marshall Space Flight Center
Presented by Rohit Sisir Sahoo

18:10 – 18:35 Featured Demo 18: Generative Geospatial User Interfaces: Dynamic Dashboard and Storytelling Generation via Agent-to-UI Streaming – EOX IT Services GmbH
Presented by Silvester Pari, Lubomír Doležal

18:35 – 19:00 Featured Demo 20: FLAP-0: An Interactive Platform for Agentic Earth Observation – Barcelona Supercomputing Center and Universitat Politècnica de Catalunya
Presented by Leonidas Kosmidis, Iker Fernandez Andonegui, Marc Solé i Bonet

Detailed Program

Day 3 - Wednesday, October 21, 2026

08:00 – 09:00 Welcome Coffee

Foyer & Wandelhalle

09:00 – 09:45 Keynote 6

Building Robust Economic Capabilities in AI Agents
Presented by Davide Crapis, Ethereum Foundation

09:45 – 10:30 Keynote 7

Why Mature Agents Require a Paradigm Shift in Tooling
Presented by Dr. Christopher Forrest Brown, Google DeepMind

10:30 – 11:00 Coffee Break & Demos

Foyer

Coffee Break

Demo 18: Generative Geospatial User Interfaces: Dynamic Dashboard and Storytelling Generation via Agent-to-UI Streaming – EOX IT Services GmbH
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Demo 19: Operating an EO Constellation with AI Agents – Aistralys
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Wandelhalle

Coffee Break

Expertisium

Coffee Break

Demo 10: BlazerOps: Reinforcement Learning Environment for Accelerated Spatial ML Development – hyades.space
Presented by Ashin Alex, Harshil Magan

Demo 16: AKD Labs: An End-to-End Agent Co-Design Studio – The University of Alabama in Huntsville, NASA Marshall Space Flight Center
Presented by Nishan Pantha, Ajinkya Kulkarni, Muthukumaran Ramasubramanian, Nidhi Jha, Rahul Ramachandran

Werner-Körte

Featured Demo 21: A Live Tour of EVE: An Agentic EO Platform Built on an Open MCP Tool Registry – Pi School
Presented by Àlex R. Atrio, Antonio Lopez, Jino Rohit, Emanuele Micieli

Rudolf-Virchow

Featured Demo 22: Agentic AI-Enabled Climate Downscaling with NASA's Prithvi WxC Foundation Model – JPL/NASA, University of Utah
Presented by Hugo Lee, Aashish Panta

11:00 – 12:00 Oral Session 4 – Multi-Modal Large Models, Datasets and Agentic Earth Intelligence *Moderated by Chair to be filled*

This session focuses on the scientific foundations that enable agentic EO systems: Vision-language models, geospatial foundation models, embeddings, semantic reasoning, multimodal grounding, and automated scientific discovery.
This session addresses: "How can foundation models become reasoning engines for Earth Intelligence?"

11:00 – 11:12 APERTURE: Semantic Change Understanding and Grounded GeoAI Reasoning via SAR–Optical–Language Alignment
Presented by Magdalena Kaiser and Paul Weinmann, LiveEO

11:12 – 11:24 BIGEARTHNET.TXT: A Large-Scale Multi-Sensor Image-Text Dataset and Benchmark for Earth Observation
Presented by Johann-Ludwig Herzog and Mathis Jürgen Adler, BIFOLD and Technische Universität Berlin

11:24 – 11:36 Toward Agentic and Unified Agricultural Field Boundary Delineation from Satellite Imagery using Geospatial Foundation Models, Pre-Trained Segmentation, and Embeddings
Presented by Sayantan Majumdar, Division of Hydrologic Sciences, Desert Research Institute

11:36 – 11:48 Evaluation of Agentic Autoresearch in an Earth Observation Benchmark
Presented by Alex López-Cifuentes, Xoople

11:48 – 12:00 GEOFMOPS: An Agentic Framework for the Geospatial Foundation Model Lifecycle in Earth Observation
Presented by Fabio Calefato, University of Bari and Nicolo' Taggio, Planetek Italia and Filippo Lanubile, University of Bari and Fabrizio Rosmarino, University of Bari

12:15 – 12:45 Closing Ceremony

12:45 – 14:00 Hosted Lunch

Langenbeck-Virchow-Haus

Food and drinks will be provided by the workshop free of charge

14:00 – 15:30 Hands-On Sessions

Expertisium: Track 1 – Building and Operating EO Agents

Building Agentic Earth Observation Workflows with EVE Workbench
Presented by Andreas Haida and Bara Safrankova, Frank Reply

B. von Langenbeck: Track 2 – Trustworthy and Scientific Agentic AI

How Do You Know Your Agent Works? Evaluating Earth Observation Agents with Existing Benchmarks
Presented by Àlex R. Atrio and Antonio Lopez and Jino Rohit, Pi School, and Nicolas Longépé, ESA Φ-Lab

Rudolf-Virchow: Track 3 – Foundation Models, Embeddings and Lightweight AI (Part 1)

Small Models, Big Maps – Local Geospatial Workflows with Fine-Tuned SLMs
Presented by Soumya Ranjan and Daniel Wiesmann, Development Seed

15:30 – 16:00 Coffee Break

Foyer & Wandelhalle

16:00 – 17:30 Hands-on Sessions

Expertisium: Track 1 – Building and Operating EO Agents

Agentic Analysis of Earth Observation for Utility Intelligence
Presented by Fuxun Yu and Rishi Madhok, TerraByte

B. von Langenbeck: Track 2 – Trustworthy and Scientific Agentic AI

From Vibe Coding to Verifiable Scientific Discovery: Agentic Workflows for Earth Observation
Presented by Rocco Sedona and Pinaki Banerjee and Alexandre Strube, Forschungszentrum Jülich, and Rohit Sahoo, The University of Alabama in Huntsville, and Àlex R. Atrio, Pi School

Rudolf-Virchow: Track 3 – Foundation Models, Embeddings and Lightweight AI (Part 2)

Small Models, Big Maps – Local Geospatial Workflows with Fine-Tuned SLMs
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